

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)		NH_4NO_3 $\frac{34}{100} \times 690 = 234.6$ $\text{CO}(\text{NH}_2)_2$ $\frac{46}{100} \times 560 = 257.6$ award (1) for one of calculations award (2) for both calculations and urea given as answer do not credit 'urea' with no working ecf possible only for minor slip in calculations		1	1	2	2	
		(ii)		ammonium nitrate is better suited to British weather conditions than urea ✓			1	1		
	(b)	(i)		either of following $(\text{NH}_4)_2\text{SO}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O} + 2\text{NH}_3$ $(\text{NH}_4)_2\text{SO}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{NH}_4\text{OH}$ products (1) balancing (1) – reactants and products must be correct for balancing mark to be awarded		2		2		2
		(ii)		$\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$ reactant ions and product (1) state symbols (1) – ions and product must be correct for state symbol mark to be awarded	1	1		2		
				Question 6 total	1	4	2	7	2	2